DisasTRACE: Emergency Incident Reporting & Dispatch Platform

Built for CDRRMO Baliwag City & Public Assistance and Command Center (PACC)

Full System Architecture, End-to-End Operational Flows, Data Models, API Registry, and Security.
Includes 1km PostGIS Spatial Dispatching, Realtime WebSockets, Touch E-Signatures, & Offline Sync.

1. EXECUTIVE SUMMARY & SYSTEM BACKGROUND

A. PROBLEM STATEMENT & BACKGROUND

• Traditional Emergency Response Challenges: CDRRMO Baliwag City previously relied on manual phone hotlines and paper-based logs. This led to verbal address miscommunications, delayed responder dispatching, lack of real-time responder location visibility, and vulnerable paper record keeping.

• Solution Overview: DisasTRACE is a centralized, multi-platform emergency management platform. It connects Public Residents, Ambulance Responders, PACC Dispatchers, and CDRRMO Executive Admins through a high-performance Android mobile app and a Next.js real-time web dashboard.

B. CORE MISSIONS & GOALS

• Eliminate Call Bottlenecks: Allow residents to submit digital SOS requests tagged with exact GPS coordinates and camera photo evidence.

• Automated 1km Proximity Selection: Use PostGIS geospatial indexing (ST_DWithin) to automatically find and dispatch the nearest available ON_DUTY ambulance within a 1.0km radius.

• Real-Time GPS Telemetry: Broadcast responder GPS positions, speed, and heading to PACC Web Command Center via Supabase Realtime WebSockets.

• Paperless Electronic Forms: Digitally capture Pre-Hospital Care Reports (PCR) and Driver's Trip Tickets (DTT) with touch finger E-Signatures and instant PDF exports.

2. STAKEHOLDER PERSONAS & SYSTEM ROLES

1. Resident / Public User (Android App): Registers with phone OTP verification and government ID. Submits emergency S

2. Ambulance Responder (Android App): Receives timed 30-second dispatch offer drawers with resident photo evidence p

3. PACC Dispatcher / Admin (Next.js Web): Monitors incoming emergency requests, triages incidents (VERIFICATION_PEN

4. CDRRMO Super Admin (Next.js Web): Reviews user/responder registrations, manages system accounts, views executiv

3. END-TO-END OPERATIONAL WORKFLOWS

A. ACCOUNT REGISTRATION & VERIFICATION GATE

• OTP SMS Verification: Users register with phone verification powered by textbee.dev SMS gateway.

• Government ID Upload: Mobile registrants upload a photo of a government-issued ID stored in Supabase Storage (ids/{userId}/).

• Super Admin Verification Gate: Accounts enter PENDING status. All mobile app screens are locked until a CDRRMO Super Admin verifies the account. If rejected, an SMS alert is sent with rejection reasons.

B. RESIDENT EMERGENCY SOS FLOW

• GPS Tagging & Photo Capture: Residents tap SOS, automatically locking GPS coordinates (latitude/longitude) and taking mandatory incident scene photos.

• Structured Classification: Captures call nature, emergency type, estimated victims, and chief complaint details.

• Live Responder Tracking: Post-dispatch, residents view assigned ambulance movement on an interactive MapLibre map with dynamic ETA updates.

C. PACC TRIAGE & VERIFICATION QUEUE

- **Verification Queue Landing: Emergency requests enter the PACC Queue as VERIFICATION_PENDING.**
- **Verification Action: Dispatchers inspect incident details, verify authenticity (VERIFIED), and choose either automated spatial dispatch or manual responder selection.**

D. AUTO-DISPATCH ENGINE & 1KM PROXIMITY CASCADE

- **PostGIS 1km Spatial Search: Executes a spatial query using ST_DWithin(location_geom, ST_SetSRID(ST_MakePoint(lng, lat), 4326), 1000) to find nearest ON_DUTY responders within 1.0km.**
- **30-Second Timed Offer: Transmits a 30-second offer (currentOfferResponderId) to the candidate. Candidate's duty status becomes ACTIVE_DISPATCH.**
- **Automated Cascading: If the responder declines or the 30-second timer expires, the engine automatically cascades to the next nearest candidate within 1km.**
- **Fallback Escalation: If no candidate accepts within 1km, the incident transitions to PACC_MANUAL status and escalates back to the PACC Dispatch Queue for manual assignment.**

E. RESPONDER MOBILE OPERATIONS & DIGITAL SIGNATURES

- **Offer Accept/Decline Drawer: Responders view timed offer cards with resident scene photos, incident location, and distance.**
- **Telemetry Broadcast: Live GPS location, heading orientation, and speed (kph) are broadcast to the command center via WebSockets (useBroadcastTracker).**
- **Turn-by-Turn Navigation & Hospital Search: Interactive route navigation to scene and receiving hospital.**
- **Touch E-Signatures (SignaturePadModal): Touch finger canvas allowing on-screen signature drawing for Patient/Guardian, Witness, Responder, Driver, and Authorized Representative.**
- **Pre-Hospital Care Report (PCR) & Driver's Trip Ticket (DTT): Electronic forms capturing vitals, GCS points, SAMPLE history, narrative reports, and vehicle fuel controls.**

F. CDRRMO EXECUTIVE DASHBOARD & PDF EXPORTS

- **Command Center Interactive Map: Displays live active incidents, hospital locations, and real-time ambulance fleet positions.**
- **KPI Analytics: Dynamic charts (Recharts) visualizing daily incidents, average response times, severity breakdown, and responder activity logs.**
- **Report Inspection & Vector PDF Export: Report inspection modal rendering SVG vector signature cards, with one-click export (exportPatientCareReportPDF & exportDriverTripTicketPDF) converting vector paths to high-res PNG images via HTML5 Canvas (convertSvgPathToDataUrl).**

G. OFFLINE DRAFT PERSISTENCE & 5-MINUTE RECURRING REMINDER LOOP

- **Local File Caching: PCR and Trip Ticket drafts created offline are saved in device storage (disas_trace_drafts.json).**
- **5-Minute System Reminder: When connection recovers (isOnline === true), the app triggers high-priority notifications and haptics every 5 minutes reminding responders to submit pending drafts.**
- **Background Queue Replay: Sequentially flushes local offline queue actions to backend API endpoints upon connection return.**

4. DETAILED TECHNOLOGY STACK SPECIFICATIONS

- **Web Front-End: Next.js 16 (React 19, App Router), TypeScript, Tailwind CSS v4, Lucide Icons, Shadcn UI, Recharts.**
- **Mobile Front-End: React Native with Expo (SDK 52, Expo Router), NativeWind / Tailwind CSS, @maplibre/maplibre-react-native (Android APK).**
- **Backend & Serverless API: Next.js API Routes (Node.js Serverless runtime), Supabase Cloud Platform.**
- **Database & Spatial Extension: Supabase PostgreSQL DBMS with PostGIS spatial extension (locationGeom spatial index, ST_DWithin 1km radius query, ST_SetSRID).**
- **ORM Layer: Drizzle ORM (type-safe SQL schema & parameterized query builder).**

- **Authentication & Security: Supabase Auth with Argon2id / bcrypt password hashing, JWT claims, 4-tier RBAC (pacc_admin, cdrrmo_admin, ambulance_responder, resident), and PostgreSQL Row Level Security (RLS).**
- **File & Asset Storage: Supabase Storage Buckets (ids/, scenes/, exports/).**
- **Mapping & Tile Engine: OpenFreeMap / MapLibre GL, OpenStreetMap (OSM) vector tiles (Zero-cost, no API key).**
- **PDF Generation: jsPDF & jsPDF-AutoTable with HTML5 Canvas vector-to-PNG renderer (convertSvgPathToDataUrl).**

5. DATABASE SCHEMA & API ENDPOINT REGISTRY

A. DATABASE TABLES (POSTGRESQL / DRIZZLE ORM)

- **public.users: Id, fullName, phone, role, dutyStatus, verificationStatus, lastLatitude, lastLongitude, locationGeom (PostGIS point).**
- **public.verification_requests: Id, residentId, locationDescription, type, latitude, longitude, imageUrl, status, createdAt.**
- **public.incidents: Id, requestId, responderId, status, assignedAmbulance, currentOfferResponderId, offerExpiresAt, dispatchMethod, createdAt.**
- **public.patient_care_reports: Id, incidentId, patientName, patientAge, dispatchInfo, emergencyType, initialAssessment, vitalsLogs, sampleHistory, narrativeReport, handoffSignatures, liabilityRelease, respondingTeam.**
- **public.driver_trip_tickets: Id, incidentId, driverName, vehiclePlate, passengerName, placesVisited, tripLog, gasolineConsumed, lubricants, speedometer, remarks, signatures.**

B. REST API ENDPOINTS REGISTRY

- **POST /api/auth/register — Mobile account registration with OTP verification.**
- **POST /api/verification-requests — Submit resident SOS help request with photo and GPS.**
- **POST /api/verification/[id]/dispatch — Manual responder dispatch by PACC dispatcher.**
- **POST /api/incidents/respond — Responder accept/decline dispatch offer.**
- **POST /api/incidents/status — Incident status updates (EN_ROUTE, ON_SCENE, TO_HOSPITAL, RESOLVED).**
- **POST /api/responder/location — Live GPS telemetry update writing PostGIS locationGeom.**
- **POST /api/reports — Submit electronic PCR and Driver's Trip Ticket payloads.**

6. SECURITY, COMPLIANCE & DISASTER RECOVERY

- **100% SQL Injection Protection: All database interactions utilize Drizzle ORM parameterized queries.**
- **Strict Role-Based Access Control (RBAC): Enforced at client navigation guards and server-side API authorization headers.**
- **Row Level Security (RLS): Supabase PostgreSQL tables are protected by strict RLS policies.**
- **Data Encryption: HTTPS / TLS 1.3 encryption for data in transit; AES-256 for storage assets.**
- **Automated Cloud Backups: Daily automated database backups with WAL Point-in-Time Recovery (PITR).**
- **Immutable Audit Trail: Every incident status transition, dispatch offer, acceptance, decline, and telemetry log is stored in audit logs.**

2026 FEATURE AND OPERATIONS UPDATE

Guest Mode now enforces a lifetime allowance by normalized phone and Android device digest.

Raw Android identifiers are never stored; the server persists only a SHA-256 digest.

Obvious repeated and sequential phone numbers are rejected at the mobile and API boundaries.

A safety notice appears before guest phone submission and explains GPS and protected device recording.

Guest status uses a scoped access token and remains available without a Supabase account session.

Automatic emergency dispatch is FIFO: the oldest confirmed emergency gets the available unit first.

Offer expiry is server-authoritative through Supabase Cron and pg_net every five seconds.

Accept is an atomic database operation and succeeds only before the server deadline.

Expired or declined offers are released, notify the responder, and remain actionable for PACC reassignment.

PACC manual dispatch uses fresh responder heartbeat, approved status, and atomic reservation checks.

Expo Push and FCM alert responders while the app is backgrounded or locked.

Push tokens are bound to the active mobile session and removed on sign-out or provider invalidation.

Native repeating reminders notify responders about unsent drafts without relying on a JavaScript timer.

Public users and responders are restricted to one active mobile device session.

Registered and guest tracking reconcile through scoped status APIs and canonical parent incidents.

Dashboard cards, charts, headings, and PACC/CDRRMO layouts are responsive and use consistent title case.

Deployment note: Play Integrity is still recommended for stronger resistance to rooted or modified clients.